American Computer Science League

2023-2024 ● Contest 1: ACSL Tiles ● Senior Division

PROBLEM: ACSL Tiles is a one-person game played with rectangular tiles. Each tile has a single-digit number, 0-9, at each end. At the start of the game, there are 4 rows, each with a single-digit number from 0-9; thus the number 405 represents the starting numbers 0, 4, 0, 5. Tiles can be re-oriented; thus, the tiles 04 and 40 are the same tile.

Tiles, called the *hand*, are given to the player, and other tiles are put into a *draw pile*. The initial number of tiles in your *hand* and in the *draw pile* varies from game to game. The goal of the game is to build rows by placing a tile from your *hand* at the right end of a row whose last number matches one of the numbers on the tile. If no tiles in your *hand* can be placed, tiles are added to the end of your *hand* from the *draw pile*, until a tile from the *draw pile* can be placed.

More specifically, on each turn, try to match your tiles starting with the first one in your *hand*. See if any can be added to one of the rows, starting with Row 1, then the row after the one where the last tile was placed, rotating back to Row 1 if necessary. However, if the last tile placed was a *double* (i.e., both numbers are the same), another tile must be placed on that row before any other match can be considered. If you cannot place any tiles in your *hand*, add tiles from the *draw pile* to your *hand* until a tile can be placed.

The game ends when you've placed all tiles in your *hand* OR you cannot place any of the tiles in your *hand* and you've exhausted the *draw pile*. At that point, find the sum of the single-digit numbers that are still in your hand.

EXAMPLE:

Input:

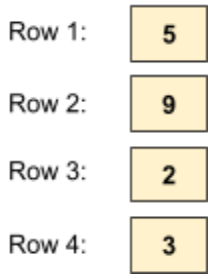
```
5923
56 27 73 34 99 45 32 17 64 57 18 11
36 92 22 50 82
```

Output:

```
16
```

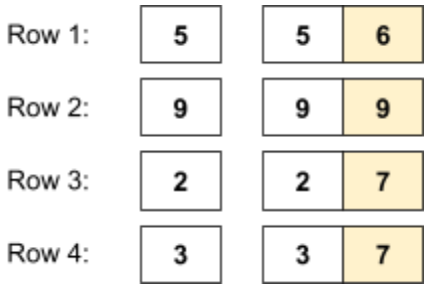
Explanation:

The game starts with 4 rows having numbers 5, 9, 2, 3.



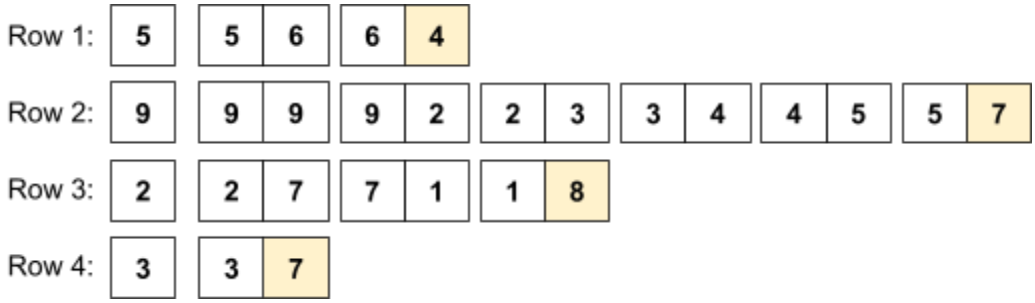
The tile 56 is placed on Row 1; the tile 27 is placed on Row 3; the tile 73 is placed on Row 4

(note that it's rotated, so that the 3 matches); the tile 34 is kept in your hand; and the tile 99 is placed on Row 2.

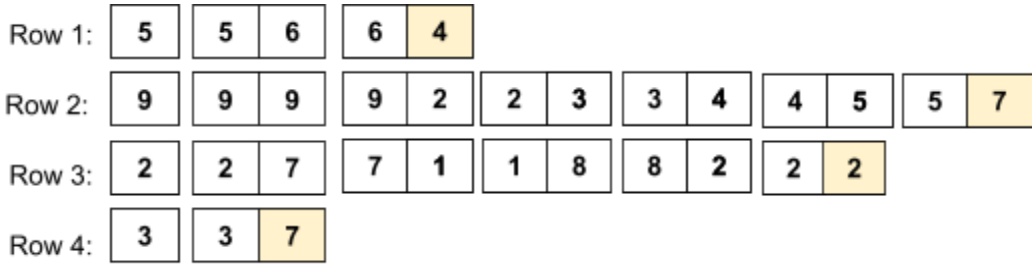


The 99 is a double, so the next tile to be placed must match a 9. None of the tiles in your hand match, so 36 is added to your hand from the draw pile. That tile cannot be placed, so it stays in your hand. Next the 92 is drawn; it can be placed on Row 2. Your hand is now 34 45 32 17 64 57 18 11 36 and the draw pile is 22 50 82.

The following moves are made: 32 on Row 2 (rotated), 34 on Row 2, 45 on Row 2, 17 on Row 3 (rotated), 64 on Row 1, 57 on Row 2, 18 on Row 3. Here is what the board looks like:



The tiles left in your hand are 11 and 36; neither can be placed, so draw 22 (it cannot be placed), then 50 (it cannot be placed), and then 82. The 82 can be placed on Row 3. Your hand is now 11 36 22 50 and the draw pile is empty. The 22 tile can be placed and then the game is over. The final board is as follows:



Your hand is 11 36 50, so the sum of the single-digit numbers on those tiles is 1+1+3+6+5+0 = 16.

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INPUT: Input an integer, between 0 and 9999 inclusive, that gives the initial numbers from Row 1 to Row 4. It is followed by two strings of numbers, from 0 to 99 inclusive, each separated by a single space. The first string has the tiles in your hand; the second string has the tiles in the draw pile. The numbers in each string represent the two numbers on each tile; a single-digit number indicates that at least one of the numbers on the tile is a 0.

OUTPUT: After placing the tiles using the rules above, output the sum of the single-digit numbers on the tiles left in your hand. If you've placed all of your tiles, then you'll output a 0.

SAMPLE INPUT:

```
1. 5923
   56 27 73 34 99 45 32 17 64 57 18 11
   36 92 22 50 82
2. 1324
   85 31 32 96 25 1 68
   30 35 42 11 78 39 19 9 81
3. 7836
   57 62 19 97 3 11 28 92 66 87 45
   68 55 58 98 38 14 53 88 44 94 81 76 74 99 27 20
4. 4
   50 0 39 98 2 99 63 46 92 74 14 58 68 33 37
   51 42 95 60 67 77 84 7 96 8 35 10 19 22 11 82 40
5. 8937
   63 84 6 57 8 2 30 9 87 52 5
   58 40 62 54 27 96 35 99 61 56 14 51 88 13
```

SAMPLE OUTPUT:

```
1. 16
2. 0
3. 102
4. 16
5. 71
```

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TEST DATA

TEST INPUT:

```
1. 1453
   24 63 52 9 85 43 6 77 19
   1 33 47 57 46 21 13 84 56 82 39 50 55 16 92 70
2. 655
   23 55 55 45 94 99 12 99 89 32
   0
3. 2594
   53 44 48 68 93 95 12 3 0 87 81 74 91 15 23 63
   72 24 83 5 76 33 26 29 7 34 8 64 16
4. 51
   84 4 70 32 58 17 38 63 51 56 27
   60 43 0 99 50 95 20 82 25 88 10 64 14 45 66 81 53
5. 0
   93 70 65 9 66 14 46 68 20 63 21 71 88 30 31 1 75
   80 84 85 47 19 89 37 26 4 76 79 92 49 51 45 53 78
```

TEST OUTPUT:

```
1. 38
2. 13
3. 11
4. 67
5. 0
```

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PROBLEM STATEMENT:

ACSL Tiles is a one-person game played with rectangular tiles. Each tile has a single-digit number, 0-9, at each end. At the start of the game, there are 4 rows, each with a single-digit number from 0-9; thus the number 405 represents the starting numbers 0, 4, 0, 5. Tiles can be re-oriented; thus, the tiles 04 and 40 are the same tile. Tiles, called the *hand*, are handed out to the placer, and other tiles are put into a *draw* pile. The initial number of tiles in your hand and in the draw pile varies from game to game. The goal of the game is to build rows by placing a tile from your hand at the right end of a row whose last number matches one of the numbers on the tile. If no tiles in your hand can be placed, tiles are added to the end of your hand from the draw pile, until a tile from the draw pile can be placed.

More specifically, on each turn, try to match your tiles starting with the first one in your hand. See if any can be added to one of the rows, starting with the row after the one where the last tile was placed, rotating back to the first row if necessary. However, if the last tile placed was a *double* (i.e., both numbers are the same), another tile must be placed on that row before any other match can be considered. If you cannot place any tiles in your hand, add tiles from the draw pile to your hand until a tile can be placed.

The game ends when you have placed all of the tiles in your hand OR you have exhausted the draw pile and cannot place any of the tiles in your hand. At that point, find the sum of the numbers that are still in your hand.

EXAMPLE:

Input:

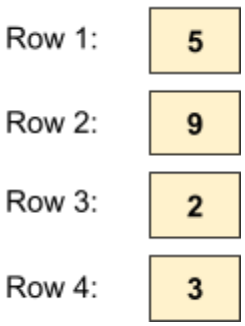
```
5923
56 27 73 34 99 45 32 17 64 57 18 11
36 92 22 50 82
```

Output:

```
16
```

Explanation:

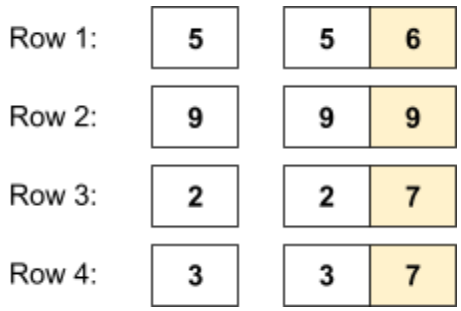
The game starts with 4 rows having numbers 5, 9, 2, 3.



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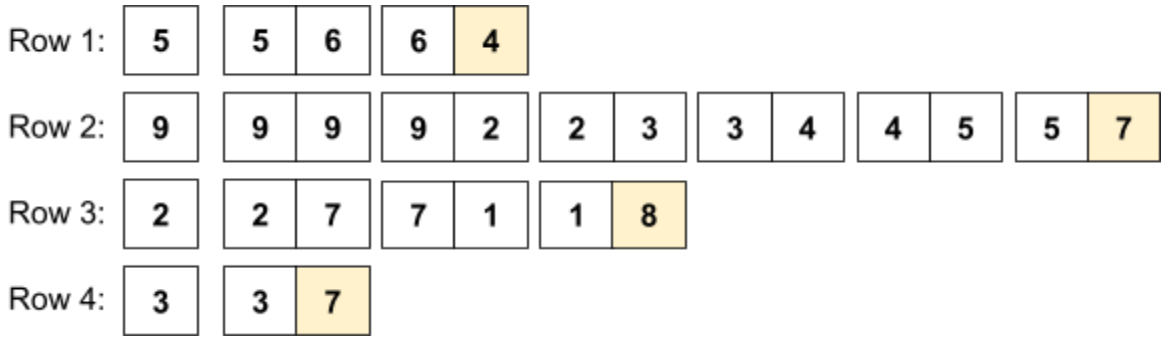
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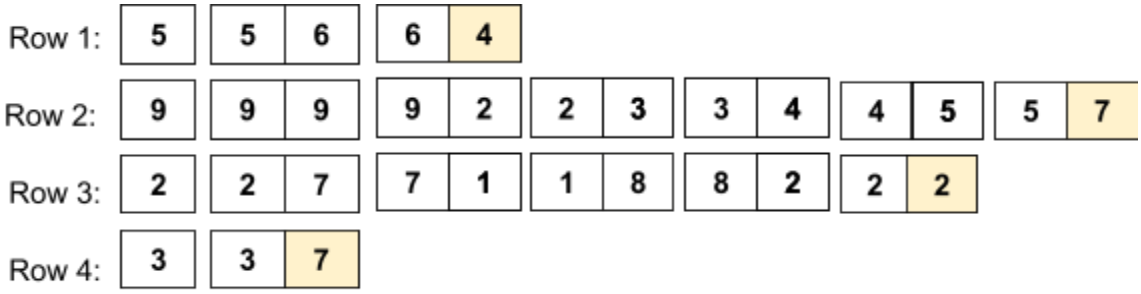


The 99 is a double, so the next tile to be placed must match a 9. None of the tiles in your hand match, so 36 is added to your hand from the draw pile. That tile cannot be placed, so it stays in your hand. Next the 92 is drawn; it can be placed on Row 2. Your hand is now 34 45 32 17 64 57 18 11 36, and the draw pile is 22 50 82.

The following moves are made: 32 on Row 2, 34 on Row 2, 45 on Row 2, 17 on Row 3 (rotated), 64 on Row 1, 57 on Row 2, 18 on Row 3. Here is what the board looks like:



The tiles left in your hand are 11 and 36; neither can be placed, so draw 22 (it cannot be placed), then 50 (it cannot be placed), and then 82. The 82 can be placed on Row 3. Your hand is now 11, 36, 22, and 50, and the draw pile is empty. Then the 22 tile can be placed, and that's it! The final board is as follows:



Your hand contains 11, 36, and 50, so the sum of those numbers is 1+1+3+6+5+0 = 16.

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TASK:

Complete the function **findHandSum**

- The function has 3 parameters: an integer, *originalRows*, between 0 and 9999 inclusive, that gives the initial numbers from Row 1 to Row 4, a string of numbers, *handTiles*, from 0 to 99 inclusive, representing the tiles in your hand, and a string of numbers, *drawPile*, from 0 to 99 inclusive, representing the tiles in the draw pile. The numbers in each string represent the two numbers on each tile; a single-digit number indicates that at least one of the numbers on the tile is a 0. The numbers in the two strings are each separated by a single space.
- The function returns an integer representing the sum of all of the single-digit numbers on the tiles that are still in your *hand* when there is nothing more that can be played or drawn.

You may create additional functions that are called from **findHandSum** if needed in solving the problem.

CONSTRAINTS:

The first input is between 0 and 9999 inclusive and all other integers are between 0 and 99 inclusive.

DATA PROVIDED:

There are 5 sets of Sample Data for debugging and 5 sets of Test Data for scoring. You may create additional data sets for debugging your program.